Module 1 exercises: dynamics of the cytokine–STAT–SOCS model

Name: _____

Use the models in the `models/ch01/` folder of the book's repository. Run them in a Jupyter notebook from the `notebooks/` folder, in VS Code, or in BNG Playground. Concentrations are in μM and time is in minutes.

A. Check the binding equilibrium

`stat_step1_binding.bngl` has `R0 = 1`, `kon = 1`, and `koff = 0.1`, so K_D = `koff`/`kon` = 0.1 μM. At equilibrium,

$[LR]_{eq} = \tfrac{1}{2}\,[\, b - \sqrt{(b^2 - 4\,L_0\,R_0)}\,]$, where $b = L_0 + R_0 + K_D$

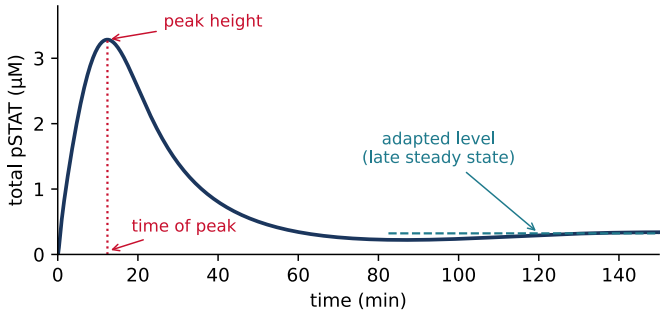
Simulate to t = 20 min so that binding reaches equilibrium (the model file's own `simulate` action stops at t = 2 min, which is too early at low ligand).

L_0 (μM)	$[LR]_{eq}$ from the formula	[LR] at the end of the simulation	Fraction of receptors bound
0.1			
1			
10			

Above roughly what ligand concentration are almost all receptors bound?

B. Tuning the pSTAT pulse

With SOCS feedback (`stat_step6_feedback.bngl`), total pSTAT rises to a peak and then falls to a lower adapted level. We describe the pulse by three properties:



The function below measures all three. The response oscillates slightly before it settles, so it simulates to t = 600 min. The same code, ready to run, is in `notebooks/pulse_exercise.ipynb`.

```
import bngsim
import numpy as np

model = bngsim.Model.from_bngl("../models/ch01/stat_step6_feedback.bngl")
sim = bngsim.Simulator(model, method="ode")

def pulse(param=None, factor=1.0):
    """Time of peak, peak height, and adapted level of total pSTAT."""
    if param is not None:
        default = model.get_param(param)
        model.set_param(param, factor * default)
    model.reset()
    res = sim.run(t_span=(0, 600), n_points=6001)
    if param is not None:
        model.set_param(param, default)        # restore
    y = res.observables["pSTAT"]
    i = np.argmax(y)
    return res.time[i], y[i], y[-1]

print(pulse())              # default parameters
print(pulse("kact", 2))     # kact doubled
```

B1. Predict before you simulate

For a **2× increase** in each parameter, predict the effect on each property. Write ↑, ↓, or — (little or no change).

Parameter	What it controls	Time of peak	Peak height	Adapted level
`L0`	ligand concentration			
`kact`	STAT activation			
`kdiss`	pSTAT release from receptor			
`kdephos`	pSTAT dephosphorylation			
`ktl`	SOCS translation			
`kSOCSdeg`	SOCS degradation			

B2. Test your predictions

Record the values from `pulse(param, 2)`. Mark any change larger than 5% with ↑ or ↓ and circle the predictions you got wrong.

Parameter doubled	Time of peak (min)	Peak height (μM)	Adapted level (μM)
none (default)			
`L0`			
`kact`			
`kdiss`			
`kdephos`			
`ktl`			
`kSOCSdeg`			

B3. Explain

1. Which parameter changes the adapted level but not the time or height of the peak? Why can it affect the late response without affecting the early one?

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2. Doubling `L0` has almost no effect. Use your answer to Part A to explain why. Now set `model.set_param("L0", 0.1)` and repeat `pulse()` and `pulse("L0", 2)`. Does ligand matter now? Why?

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3. A drug should lower the adapted response while leaving the initial peak intact. Which of the parameters in your table would you target, and in which direction?

C. Remove the phosphatase

In `stat_step4_release.bngl` (no SOCS yet), set `kdephos` to 0.

Before simulating, predict the value of total pSTAT after a long time:

After simulating: what sets this value, and why is there no longer a balance between activation and deactivation?

D. Toward Module 2

In our model, SOCS can bind only the empty ligand-bound receptor LR. Suppose SOCS can also bind receptors that carry unphosphorylated STAT (LRS).

Write the new reaction in the notation of Module 1, and name the new species.

What other reactions does the new complex need (for example, can STAT still unbind from it, or be activated)? List them.

Exit response

1. In one sentence: why does negative feedback produce a peak instead of a steady rise to a plateau?

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2. Which steps in the model create the delay before the feedback takes effect?
